INTERNSHIP EVALUATION · 2026

DarkTrace X Chatbot

A conceptual AI assistant for cybersecurity and digital-threat awareness

PRESENTED BY

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UNIVERSITY

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PROGRAM & SEMESTER

B.Tech CSE
Semester 7

ORGANIZATION

Diligent Global Tech Consulting Pvt. Ltd.

01 / INTERNSHIP CONTEXT

A structured internship in applied AI

The project connected classroom learning in computer science with the design of an AI-enabled cybersecurity-awareness concept.

08 JUN — 31 AUG 2026 Internship programme duration

ROLE

AI Intern

DEPARTMENT

Technology and AI

ORGANIZATION

Diligent Global Tech Consulting Pvt. Ltd.

LOCATION

Hitech City, Hyderabad, India

PERFORMANCE

OUTSTANDING

ACADEMIC CONTEXT

B.Tech CSE · Semester 7

Why cybersecurity awareness matters

THE USER REALITY

Suspicious digital activity often reaches users before they understand the relevant security concepts.

THE DESIGN OPPORTUNITY

01 Phishing and social engineering arrive through many channels.

02 Guidance is often technical, scattered, or hard to apply quickly.

03 Users need clear, cautious, and actionable support.

04 Good assistance should reduce uncertainty—not hide it.

03 / PROBLEM STATEMENT

The Usability and Awareness Gap

Technical Knowledge

Security guidance is often dispersed across policy pages, technical advisories, and specialist tools.

User Application

Users must make split-second decisions when facing suspicious messages, links, or login requests.

CORE CHALLENGE

"A gap exists between the availability of security knowledge and the user’s ability to apply it safely in context."

04 / THE SOLUTION

DarkTrace X: A conceptual AI assistant

INTERFACE

NATURAL LANGUAGE INPUT

Accepts free-form descriptions of suspicious messages, links, login requests, and device behavior.

INTELLIGENCE

AWARENESS CATEGORIZATION

Identifies relevant indicators and maps them to broad threat categories like phishing or impersonation.

GUIDANCE

CAUTIOUS EXPLANATION

Provides safe next steps and escalation paths using uncertainty-aware language to avoid false certainty.

05 / INTERACTION MODEL

From User Observation to Safe Action

STAGE 01	STAGE 02	STAGE 03	STAGE 04	STAGE 05
DESCRIBE	**CLARIFY**	**INTERPRET**	**GUIDE**	**ESCALATE**
Capture the user's observation in free-form natural language without technical constraints.	Ask focused questions when the initial context is insufficient for a reliable assessment.	Map the description to awareness categories and identify specific threat indicators.	Explain safe next steps using uncertainty-aware language to avoid false certainty.	Recommend human or organizational support for serious cases or suspected compromise.

Core awareness categories

CATEGORY	COMMON INDICATORS	AWARENESS RESPONSE DIRECTION
Phishing	Unexpected request, urgency, suspicious link, impersonated sender.	Avoid the link, verify independently, report the message.
Credential Theft	Request for password, MFA code, or account recovery details.	Do not disclose secrets; change affected credentials if exposed.
Malware Delivery	Unexpected attachment, download prompt, or executable file.	Do not open; scan the device and contact support if opened.
Impersonation	Brand or person identity appears inconsistent or unusually urgent.	Confirm through an official channel before acting.
Account Compromise	Unrecognised login, password reset, or security notification.	Secure the account, review activity, and escalate if necessary.

07 / SYSTEM DESIGN

Proposed logical architecture

USER INTERFACE	Collects **minimum context** and displays privacy warnings to prevent secret disclosure.
INPUT PROCESSING	Validates text, normalizes language, and identifies **intents and entities**.
AI INTELLIGENCE	Interprets the situation and prepares a **response plan** using least-privilege principles.
THREAT KNOWLEDGE	Connects to **approved, versioned** awareness guidance and validated indicators.
RESPONSE GENERATION	Produces explanations and safe actions while applying **refusal and validation** rules.
HUMAN OVERSIGHT	Escalates **suspected compromise** or high-impact cases to qualified security support.

Proposed modular components

CORE LANGUAGE

PYTHON

Modular processing, orchestration, validation, and logging for the conceptual assistant.

LANGUAGE PROCESSING

NLP PIPELINE

Normalization, intent classification, entity extraction, and context handling.

INTELLIGENCE

GENERATIVE AI

Drafting explanations with prompt controls, grounding, and refusal behavior.

INTERACTION

CHATBOT FRAMEWORK

Dialogue state management, routing, clarification questions, and session handling.

INFRASTRUCTURE

BACKEND / API

Connecting the interface, processing services, and governed knowledge sources.

KNOWLEDGE

AWARENESS DATA

Approved, versioned cybersecurity guidance and validated threat indicators.

Note: All components represent proposed design choices for the conceptual project and do not imply production implementation.

Responsible AI and Privacy by Design

PRIVACY FILTERS

Discourages disclosure of **passwords, keys, and MFA codes**. Sensitive data is masked or rejected at the interface layer.

UNCERTAINTY MANAGEMENT

Separates observations from interpretations. Uses **cautious language** to avoid creating false certainty in users.

OUTPUT VALIDATION

Checks generated responses against **deterministic safety rules** to prevent unsafe operational instructions.

HUMAN ESCALATION

Routes suspected compromise or high-impact cases to **qualified human support** rather than taking autonomous action.

Academic workflow and evaluation

NINE-STAGE PROJECT METHODOLOGY

01 Requirement Analysis

02 Problem Identification

03 System Design

04 Conversational Design

05 AI Intelligence Design

06 Threat-Awareness Logic

07 Interface Design

08 Conceptual Evaluation

09 Documentation

CONCEPTUAL EVALUATION CRITERIA

- **Clarity**
- **Relevance**
- **Safety**
- **Privacy**
- **Consistency**
- **Uncertainty**

Outcomes and Future Directions

EXPECTED OUTCOMES

- Improved user interaction through natural-language processing.
- Structured guidance separating observations from safe actions.
- Foundation for integration with validated security knowledge.

CONCEPTUAL LIMITATIONS

- Documents a conceptual design, not a verified deployment.
- Cannot guarantee safety from incomplete user information.

FUTURE SCOPE

- Develop controlled prototypes for intent and entity extraction.
- Scenario-based evaluation with human security reviewers.
- Study multilingual support and accessibility requirements.
- Explore privacy-preserving logging and audit trails.

Thank You.

SELECTED REFERENCES

[1] NIST, Phishing Guidance and Awareness Resources

[2] NIST, Digital Identity Guidelines

[3] OWASP, Top 10 Risks for LLM Applications

[4] Python Software Foundation / Streamlit Documentation